ESE 2113 / MODULE 03

# Measures of Central Tendency

Structured teaching edition: concepts, calculations and interpretation

Second Year, First Semester

# Learning goals and prerequisites

**Prerequisite: Ordered observations, frequencies and midpoints.**

By the end: Calculate several measures of centre. Match weighting to the scientific question. Explain sensitivity to extreme observations.

Study route: definitions, explicit calculations, visual interpretation, then independent practice and solutions. All numerical examples are illustrative unless a source says otherwise.

# Notation used in this lecture

| Symbol | Meaning |
|---|---|
| $\bar{x}$, $\mu$ | Sample and population arithmetic means |
| $n$, $N$ | Sample and population sizes |
| $w_i$ | Weight assigned by the study question |
| $G$, $H$ | Geometric and harmonic means |
| $L$, $h$ | Lower boundary and class width |
| $\Sigma$, $\prod$ | Add terms and multiply terms |

Read the symbol as a quantity with a meaning, not merely as a letter to substitute.

# Arithmetic mean, median and mode

## Part 1 of 3

The definitions and calculations in this part prepare you to interpret its graphs.

1. Central tendency: definition and purpose
2. Arithmetic mean
3. Why the mean is a balance point
4. Median
5. Mode and modal class

# Central tendency: definition and purpose

**A measure of central tendency represents the location or typical level of a distribution using one value. Different definitions summarize different aspects.**

Arithmetic mean uses magnitudes, median uses ordered position, and mode uses frequency. The scientific question determines the useful choice.

We use 2,3,3,4,18 µg/L repeatedly: a small synthetic sample with one high measurement.

# Arithmetic mean

**For n values $x_1,\ldots,x_n$, $\bar{x}=\Sigma x_i/n$. The population mean uses all N population values and is denoted $\mu$.**

The mean uses every numerical value. Increasing one observation by d increases the mean by d/n.

The deviations balance: $\Sigma(x_i-\bar{x})=0$. This property explains the mean as a balance point.

For 2,3,3,4,18 µg/L, sum=30 and n=5, so mean=6 µg/L.

# Why the mean is a balance point

**For 2,3,3,4,18, the sum is 30 and the mean is 6. Deviations are −4,−3,−3,−2,+12.**

Their sum is zero. The four values below 6 balance the contribution of the high observation.

Algebra: $\Sigma(x_i - \bar{x}) = \Sigma x_i - n\bar{x} = \Sigma x_i - n(\Sigma x_i / n) = 0$.

The mean need not be an observed value or the most common result.

# Median

**After sorting, the median is the middle value for odd n and the average of the two middle values for even n.**

For 2,3,3,4,18, the median is 3. For 2,3,4,18, it is (3+4)/2=3.5.

The median is resistant to a few extreme observations. It does not summarize the total mass or total load.

A typical household exposure may suit a median, while total contaminant loading requires an appropriate mean or integral.

# Mode and modal class

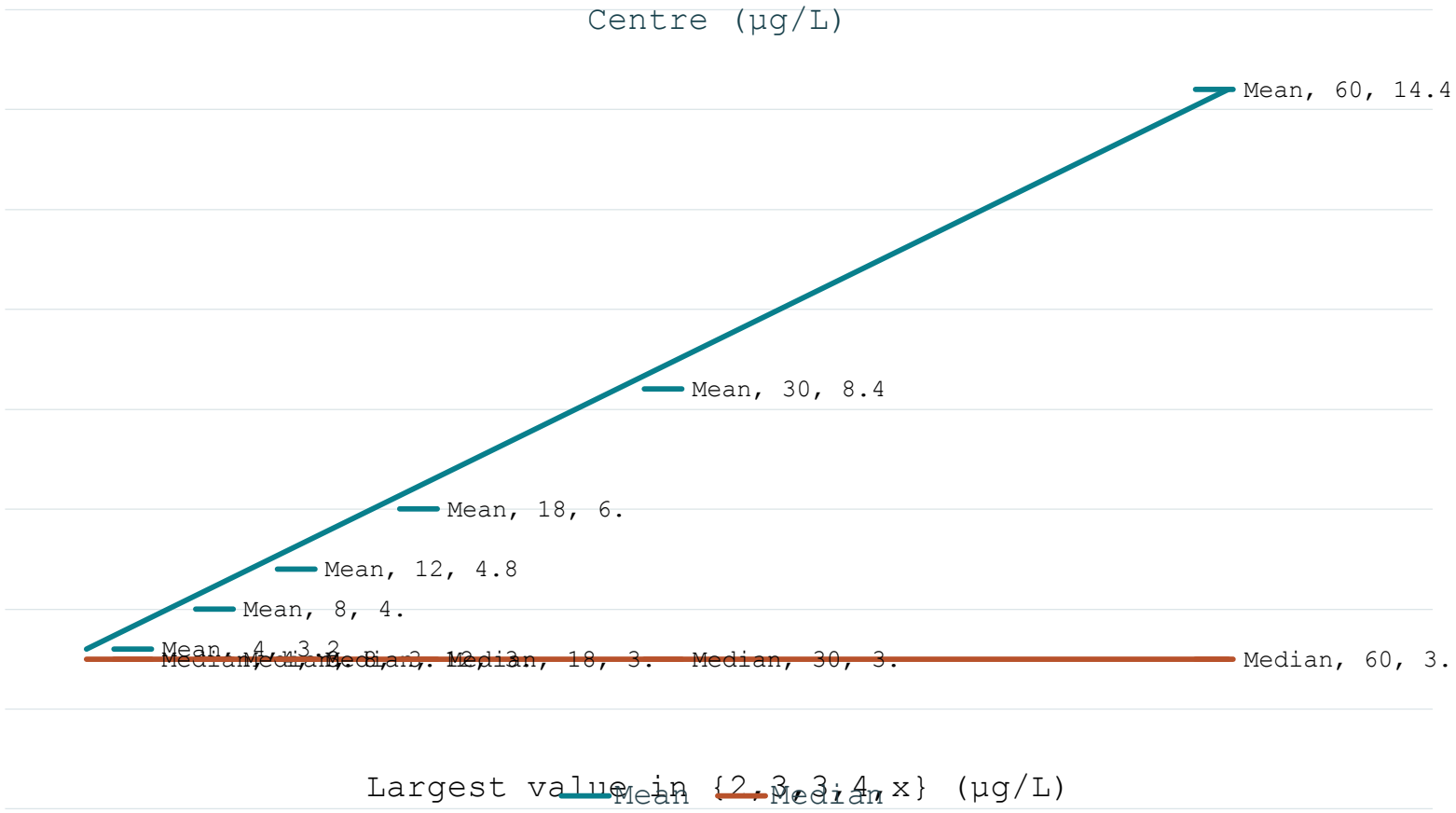
**The mode is the most frequent observed value. A sample can have several modes or no unique mode.**

For 2,3,3,4,18 the mode is 3. For continuous measurements, exact repetitions can depend on instrument rounding.

The modal class is the most frequent histogram class, which may change when boundaries change.

A categorical mode is meaningful even when a numerical mean is not, such as the most frequent land-use class.

# An extreme value moves the mean much more




## How to read this graph

Four observations remain unchanged.

The mean responds to every increase in x.

The median stays at 3 throughout this example.

Illustrative data or theoretical distribution. Exact plotted values remain editable.

# Checkpoint: Sensitivity of the mean

**Question: Replace 18 by 28 in 2,3,3,4,18. What changes?**

Worked answer: The total increases by 10, so the mean increases by 10/5=2, from 6 to 8. Median=3 and mode=3 remain unchanged.

# Weighted, geometric and harmonic means

## Part 2 of 3

The definitions and calculations in this part prepare you to interpret its graphs.

1. Weights express how observations contribute
2. Weighted arithmetic mean
3. Worked flow-weighted concentration
4. Geometric mean
5. Why a geometric mean uses logarithms
6. Harmonic mean

# Weights express how observations contribute

**A weighted mean gives each value a stated contribution. The weights must follow the estimand: duration, volume, flow or number of represented units.**

If concentration is 10 for 2 hours and 30 for 6 hours, exposure total is $10 \times 2 + 30 \times 6 = 200$ concentration-hours.

Divide by total time 8 hours to get 25. The denominator cancels the weight unit, leaving concentration.

# Weighted arithmetic mean

**For nonnegative weights $w_i$, $\bar{x}_w = \Sigma w_i x_i / \Sigma w_i$. Weights reflect exposure duration, group size or sampling probability, not personal preference.**

Two hours at 10 µg/m³ and six hours at 30 give $(2\times10+6\times30)/8=25$ µg/m³.

The unweighted mean 20 would give each period equal importance despite unequal duration.

For flow-weighted water concentration, use $\Sigma Q_i C_i / \Sigma Q_i$ when the time intervals are equal.

# Worked flow-weighted concentration

| Interval | Flow (L/s) | Concentration (mg/L) | Q × C (mg/s) |
|---|---|---|---|
| 1 | 2 | 5 | 10 |
| 2 | 8 | 15 | 120 |
| Sum | 10 | — | 130 |

Flow-weighted concentration =130/10=13 mg/L. Unweighted mean=10 mg/L. Intervals have equal duration.

# Geometric mean

**For positive values, $G=(\prod x_i)^{(1/n)}=\exp[\Sigma \ln(x_i)/n]$. It summarizes multiplicative rather than additive structure.**

For 1,10,100, G=(1000)^(1/3)=10, while the arithmetic mean is 37.

For lognormal data, the geometric mean estimates the distribution's median, not its arithmetic mean.

Zeros, negative values and nondetects require explicit treatment. Adding 1 changes the estimand and must be justified.

# Why a geometric mean uses logarithms

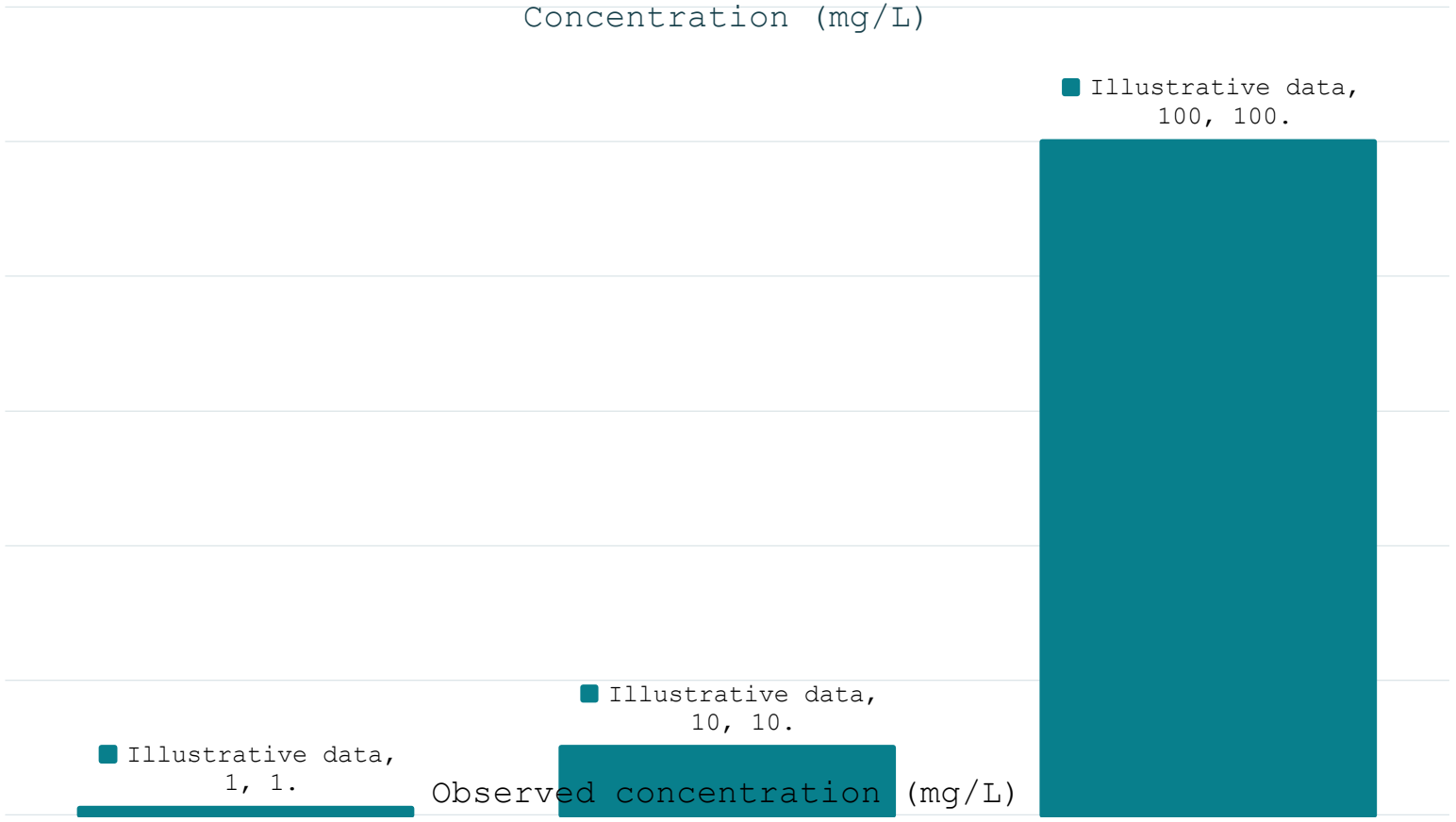
**For positive values 1,10,100, log10 values are 0,1,2. Their arithmetic mean is 1. Back-transform: $10^1 = 10$.**

The product approach gives (1×10×100)^(1/3) = 10, the same answer.

Equal log distances represent equal ratios. This makes the geometric mean useful for multiplicative patterns.

The arithmetic mean is 37. It answers an additive question and cannot be replaced by 10 when estimating total mass in equal-volume samples.

# Arithmetic versus geometric mean




Illustrative data or theoretical distribution. Exact plotted values remain editable.

## How to read this graph

Arithmetic mean =37 mg/L.

Geometric mean =10 mg/L.

The appropriate summary depends on whether the question is additive or multiplicative.

# Harmonic mean

**For positive values, $H=n/\Sigma(1/x_i)$. It applies to rates when equal amounts of the numerator are combined.**

Travel over equal distances at 20 and 60 km/h gives $H=2/(1/20+1/60)=30$ km/h.

For equal time spent at those speeds, the correct average is the arithmetic mean 40 km/h.

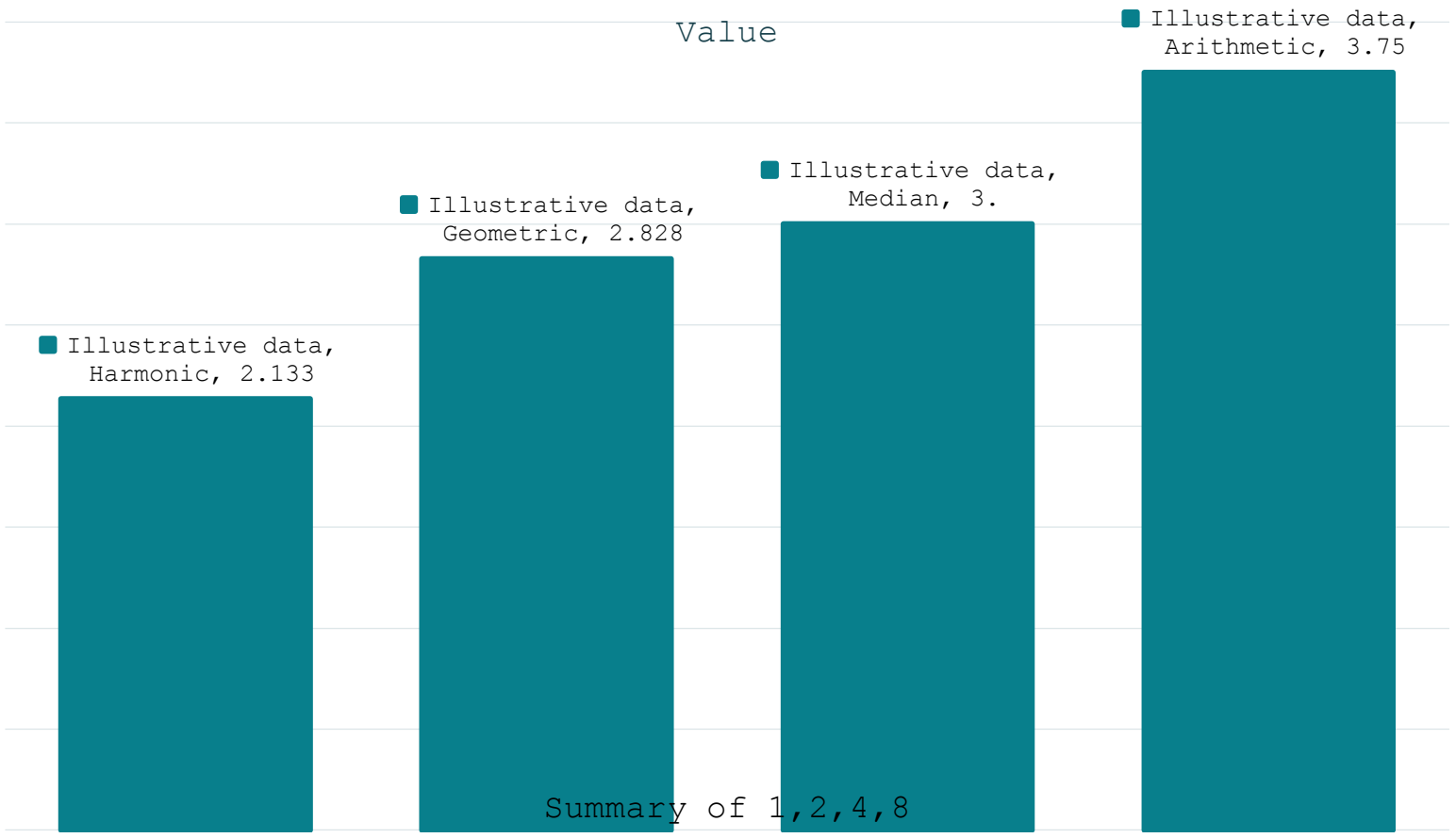
For positive observations, $H \leq G \leq A$. Equality holds when all observations are equal.

# One dataset, several valid centres

| Measure | Data: 1,2,4,8 | What it emphasizes |
|---|---|---|
| Arithmetic | 15/4=3.75 | Additive magnitude |
| Median | (2+4)/2=3 | Middle position |
| Geometric | 64^(1/4)=2.8284 | Multiplicative structure |
| Harmonic | 4/1.875=2.1333 | Rates under appropriate weighting |

All values are positive. H≤G≤A holds. Harmonic denominator=1+1/2+1/4+1/8. Choose the average for the question.

# Comparison of four centres




Illustrative data or theoretical distribution. Exact plotted values remain editable.

## How to read this graph

Different definitions give different numbers from the same data.

Neither the largest nor the smallest is automatically the best.

For total mass per equal-volume sample, arithmetic mean has a direct additive interpretation.

# Combining subgroup means

**If group j has nj observations and mean $\bar{x}_j$, the pooled mean is $\Sigma n_j \bar{x}_j / \Sigma n_j$.**

Suppose 5 upstream samples average 2 mg/L and 15 downstream samples average 6 mg/L. Pooled mean=(5×2+15×6)/20=5 mg/L.

A simple average of the two means would be 4 and would weight the locations equally, not the observations.

Either weighting could target a different question. State whether the estimand is the mean sampled observation or an equally weighted location mean.

# Checkpoint: Choosing weights

**Question: Two equal-duration periods have flows 2 and 8 and concentrations 5 and 15. Find the flow-weighted concentration.**

Worked answer: Weighted total=2×5+8×15=130. Total flow weight=10, so the mean is 13. An unweighted mean of 10 treats low- and high-flow periods equally and answers a different question.

# Grouped data and robust summaries

## Part 3 of 3

The definitions and calculations in this part prepare you to interpret its graphs.

1. A grouped average replaces unknown positions
2. Grouped mean
3. Derivation of the grouped median formula
4. Grouped median and mode
5. Trimmed mean and robust summaries

# A grouped average replaces unknown positions

**A grouped frequency table retains counts but loses exact values inside each interval. Any midpoint calculation therefore makes an approximation.**

For intervals [0,10),[10,20),[20,30), use representatives 5,15,25. With counts 2,5,3, the weighted total is 160.

Dividing by 10 observations gives an estimated mean of 16. Actual observations near class edges could give a different exact mean.

# Grouped mean

**When only class counts are available, use $\bar{x} \approx \Sigma f_i m_i / n$, with midpoint $m_i$.**

For classes [0,10), [10,20), [20,30) with frequencies 2,5,3: midpoints are 5,15,25.

$\Sigma fm = 2 \times 5 + 5 \times 15 + 3 \times 25 = 160$, giving grouped mean=16.

This is approximate because individual values need not equal their class midpoints.

# Derivation of the grouped median formula

**The median lies halfway through the sample. Find the class whose cumulative frequency reaches n/2.**

If L is its lower boundary, Fprev observations lie below it and f observations occupy width h, the fraction needed inside the class is (n/2−Fprev)/f.

Estimated distance into the class = h×(n/2−Fprev)/f. Add L to obtain the grouped median.

This linear interpolation treats observations as approximately evenly spread inside the median class.

# Grouped median and mode

**Grouped median ≈ L + [(n/2−Fprevious)/fmedian]h. L is the lower median-class boundary and h its width.**

With frequencies 2,5,3 in width-10 classes, median≈10+[(5−2)/5]10=16.

Grouped mode≈L+[(fm−fprevious)/(2fm−fprevious−fnext)]h for an interior modal class of equal width.

Here mode≈10+(3/5)10=16. These interpolation formulas assume smooth within-class behavior.

# Trimmed mean and robust summaries

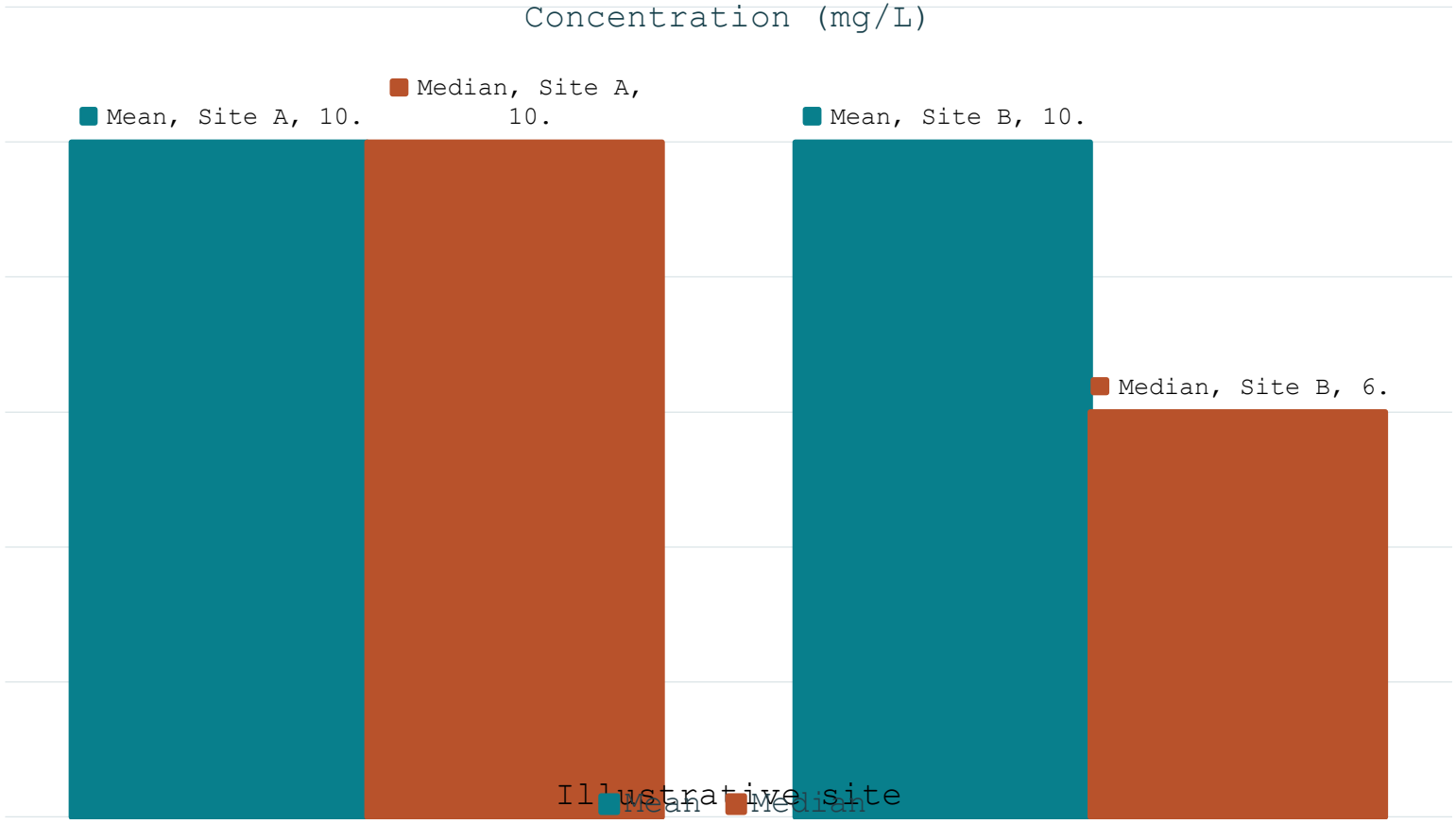
**A trimmed mean removes a stated fraction from each tail before averaging. A 10% trim removes floor(0.1n) observations from each side in R.**

Trimming limits extreme-value influence but changes the target summary. It is not a licence to remove inconvenient pollution spikes.

A winsorized mean replaces extremes with boundary values rather than deleting them. State which approach you used.

Always retain raw data and explain whether extremes are valid environmental events or confirmed errors.

# Site summaries answer different questions




Illustrative data or theoretical distribution. Exact plotted values remain editable.

## How to read this graph

Site A: 8,9,10,11,12.
Site B: 1,2,6,11,30.

Both means are 10, but the medians are 10 and 6.

A single centre statistic hides the high value at Site B.

# Checkpoint: Grouped approximation

**Question: Classes [0,10),[10,20),[20,30) contain 2,5,3 values. Why is their grouped mean 16 approximate?**

Worked answer: It assumes each value equals its class midpoint: (2×5+5×15+3×25)/10=16. Different raw values can have the same frequencies but different exact means.

# Practice, solutions and reproducible analysis

## Apply the complete method

Attempt each problem before revealing its solution. Show units and explain the conclusion.

1. Choosing and calculating an average: practice
2. Choosing and calculating an average: worked solution
3. Median from a grouped distribution: practice
4. Median from a grouped distribution: worked solution
5. R laboratory: reproducible analysis
6. Excel laboratory: formulas and interpretation

# Choosing and calculating an average: practice

**Concentrations 4 and 16 mg/L are measured for 1 h and 3 h. Find the time-weighted mean.**

Find the geometric mean of 4 and 16.

Can the mean of pH values be interpreted as the mean hydrogen-ion concentration?

# Choosing and calculating an average: worked solution

**Time-weighted mean=(4×1+16×3)/4=13 mg/L.**

Geometric mean=√64=8 mg/L.

No. pH is logarithmic. If the objective is mean $H^+$ concentration, transform to 10^(−pH), average, then interpret appropriately.

# Median from a grouped distribution: practice

**Classes [0,10),[10,20),[20,30),[30,40) have frequencies 3,7,8,2.**

Estimate the mean and median. Explain the approximation.

# Median from a grouped distribution: worked solution

**Midpoint total=3×5+7×15+8×25+2×35=390. Grouped mean=390/20=19.5.**

The 10th observation lies at the cumulative boundary 20. Interpolation gives median≈20.

Individual within-class positions are unknown. Grouped centre estimates cannot reproduce every possible raw dataset.

# R laboratory: reproducible analysis

```
x <- c(2,3,3,4,18)
mean(x); median(x)
exp(mean(log(x)))
weighted.mean(c(4,16),c(1,3))
```

Mean=6, median=3, mode=3. The time-weighted example returns 13 mg/L.

# Excel laboratory: formulas and interpretation

**A2:A6 contains 2,3,3,4,18. =AVERAGE(A2:A6), =MEDIAN(A2:A6), =MODE.SNGL(A2:A6).**

=GEOMEAN(A2:A6) requires positive values.

=SUMPRODUCT(B2:B3,C2:C3)/SUM(C2:C3) calculates a weighted mean for values and weights.

Mean=6, median=3, mode=3. The time-weighted example returns 13 mg/L.

# A connected review of this lecture

**Arithmetic mean, median and mode: A measure of central tendency represents the location or typical level of a distribution using one value. Different definitions summarize different aspects.**

Weighted, geometric and harmonic means: A weighted mean gives each value a stated contribution. The weights must follow the estimand: duration, volume, flow or number of represented units.

Grouped data and robust summaries: A grouped frequency table retains counts but loses exact values inside each interval. Any midpoint calculation therefore makes an approximation.

# Reading pathway and mathematical extensions

**Core reading: OpenStax §2.5; NIST EDA §1.3.5.1; EPA QA/G-9S descriptive statistics**

MIT 18.05: use the topic-matched readings, in-class problems and solutions listed in the course guide. MIT material supplements these original ESE explanations.

Mathematical extension: Hogg & Craig or Mood, Graybill & Boes for derivations. Applied worked practice: Spiegel. Use the topic index for your available edition.

These lectures are independent teaching materials, not MIT course slides or an MIT-certified curriculum.

# Open-access reading links

https://ocw.mit.edu/courses/18-05-introduction-to-probability-and-statistics-spring-2022/pages/classes-reading-and-in-class-materials/

OpenStax: https://openstax.org/books/introductory-statistics-2e/pages/1-introduction

NIST: https://www.itl.nist.gov/div898/handbook/

EPA: https://www.epa.gov/quality/guidance-data-quality-assessment

Penn State: https://online.stat.psu.edu/stat414/

R: https://stat.ethz.ch/R-manual/R-devel/library/stats/html/00Index.html

Excel: https://support.microsoft.com/en-us/excel/statistical-functions-reference